UNIVERSAL MOTHER-WAVELET DECOMPOSITION OF THE JACOBI–RIEMANN KERNEL, SMOOTH JACOBI–SINH SHELLS, AND EXPLICIT SADDLE-POINT REMAINDER BOUNDS FOR RIEMANN'S Ξ-FUNCTION

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Abstract. We establish the harmonic and saddle-point decomposition of the Jacobi–Riemann kernel $\Phi(u) = \sum_{m=1}^{\infty} \phi_m(u)$ governing the transverse Wronskian $W(d,t) = RI_t - IR_t$ of Riemann's $\Xi(t - \mathrm{i}d) = R + \mathrm{i}I$. Every shell $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$ is an exact logarithmic translation of a single mother wavelet $\varphi_1(w)$ whose bilateral Laplace–Mellin transform $\mathcal{G}(s) = \frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(s/2)$ exhibits exact polynomial cancellation $|(1-s)(-s)| \equiv |s(s-1)|$ along $s = \frac{1}{2} + d + \mathrm{i}t$, yielding a three-line reflection-ellipse proof of $W_{1,1}(d,t) > 0$ and the Symmetric Branch-Damping Inequality across all active Riemann–Siegel modes $1 \le n \le N(t) = \lfloor\sqrt{t/2\pi}\rfloor$. Smooth self-mirrored Jacobi–Sinh shells eliminate the $u = 0$ odd-derivative kink of naive half-line truncations, while the diagonal sum $\mathcal{D}_{\mathrm{diag}}(d,t)$ obeys a non-oscillatory $(\ln t)^3$ Stieltjes expansion that dominates both the $\mathcal{O}(t^{-3/2})$ self-damped boundary cross-pair $(N-1, N)$ and explicit incomplete-Gamma and two-dimensional Gabcke–Olver remainder envelopes.

1. Introduction and Main Results

In his foundational 1859 memoir, Riemann [13] introduced the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and its critical-line rotation $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$, represented as the Fourier cosine transform of the super-exponentially decaying Jacobi theta kernel [5, 12, 17]. Decomposing $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}\,I(d,t)$ for spectral height $t \in \mathbb{R}$ and transverse displacement $d = \sigma - \frac{1}{2} > 0$, the transverse Wronskian $W(d,t) := R(d,t)I_t(d,t) - I(d,t)R_t(d,t) = |\Xi(t - \mathrm{i}d)|^2 \partial_t \arg \Xi(t - \mathrm{i}d)$ governs both the vertical monotonicity of $|\Xi(t - \mathrm{i}d)|^2$ via the Cauchy–Riemann identity $\partial_d(|\Xi|^2) = 2W(d,t)$ and the topological zero-detection criteria of Speiser [15], Hinkkanen [3], and Lagarias [6].

In Papers I–IV of this series [7, 8, 9, 10], the Wronskian $W(d,t)$ was cast into Wigner phase-space coordinates $(x,y) = (u+v, u-v)$ [18, 4] and analyzed through complex saddle-point coalescence, parity-reduced kernel stiffness, Chladni nodal chamber geometry, and de Branges–Jensen positivity [1]. A central structural problem remains at the bridge between the exact integral representation of $\Xi(z)$ and the finite Riemann–Siegel Dirichlet sum [14, 2, 11] of length $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$: how do the individual summation modes $\phi_m(u)$ of the Jacobi–Riemann kernel $\Phi(u) = \sum_{m=1}^{\infty} \phi_m(u)$ interact in the bilinear Wronskian matrix, why does naive half-line shell splitting produce a severe $\mathcal{O}(t^{-2})\mathrm{e}^{\pi t/2}$ boundary artifact at $u = 0$, why does the slowest adjacent boundary pair $(N-1, N)$ not destabilize the diagonal sum as its frequency gap $\Omega_{\min}(t) = \ln(N/(N-1)) \sim \sqrt{2\pi/t}$ narrows, and how are the infinite saddle-point tails bounded explicitly across the two-dimensional half-strip $\mathcal{S}^+_{1/2} := \{(d,t) \in (0, 1/2] \times [14, \infty)\}$?

The present paper (Paper V) resolves these four questions within a unified analytical framework. Our main results are consolidated in the following theorem.

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Theorem 1.1 (Universal Mother-Wavelet Decomposition, Smooth Jacobi–Sinh Shells, and Explicit Remainder Bounds)**.** *Let* $\Phi(u) = \sum_{m=1}^{\infty} \phi_m(u)$ *with* $\phi_m(u) = (2\pi^2 m^4 \mathrm{e}^{9u} - 3\pi m^2 \mathrm{e}^{5u}) \exp(-\pi m^2 \mathrm{e}^{4u})$, *and let* $s = \frac{1}{2} + d + \mathrm{i}t$ *with* $z = t - \mathrm{i}d$. *Then the following four structural laws hold:*

(i) ***Universal Mother-Wavelet Dilation and Symmetric Branch Damping.*** *Every shell* $\phi_m(u)$ *is an exact logarithmic translation of the mother wavelet* $\varphi_1(w) := \phi_1(w)$*:*

$$\phi_m(u) = m^{-1/2}\, \varphi_1\!\left(u + \frac{1}{2}\ln m\right) \qquad (m \geq 1,\ u \in \mathbb{C}), \tag{1.1}$$

with bilateral Laplace–Mellin transform $\mathcal{G}(s) := \int_{-\infty}^{\infty} \varphi_1(v) \mathrm{e}^{2(s-1/2)v}\, \mathrm{d}v = \frac{1}{8} s(s-1)\pi^{-s/2}\Gamma(s/2)$. *Because* $|(1-s)(-s)| \equiv |s(s-1)|$ *holds identically on* $\mathbb{C}$*, the polynomial prefactor cancels completely in the reflection modulus ratio* $\rho_\Gamma(d,t) := |\mathcal{G}(1-s)/\mathcal{G}(s)| = \pi^d |\Gamma(\frac{1}{4} - \frac{d}{2} + \mathrm{i}\frac{t}{2})/\Gamma(\frac{1}{4} + \frac{d}{2} + \mathrm{i}\frac{t}{2})| = \exp(-2\int_0^d \mathcal{A}_\Gamma^{\mathrm{sym}}(u,t)\,\mathrm{d}u) < 1$ *for all* $d > 0$ *and* $t \geq 7 > t_{\mathrm{crit}} = 6.2898359888\ldots$. *Consequently,* $W_{1,1}(d,t) > 0$ *and every active mode* $1 \leq n \leq N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ *obeys the Symmetric Branch-Damping Inequality:*

$$\rho_\Gamma(d,t)\, n^{-1/2+d} < \mathrm{e}^{-2d\,\omega_n(t)}\, n^{-1/2-d} \leq n^{-1/2-d}, \qquad \omega_n(t) := \vartheta'(t) - \ln n \geq 0. \tag{1.2}$$

(ii) ***Wigner Phase-Space Shift and Kink-Free Jacobi–Sinh Shells.*** *In Wigner coordinates* $(x,y) = (u+v, u-v)$*, the cross-shell product* $\phi_m(u)\phi_n(v)$ *translates the mother-wavelet kernel to* $(x_{m,n}, y_{m,n}) = (-\frac{1}{2}\ln(mn), \frac{1}{2}\ln(n/m))$ *with strictly positive Dirichlet–Sinh lever* $\mathcal{L}_{m,n}(d) := y_{m,n}\sinh(d y_{m,n}) = \frac{1}{4}\ln(n/m)[(n/m)^{d/2} - (m/n)^{d/2}] > 0$ *for all* $m \neq n$ *and* $d > 0$*. While naive half-line shells* $\phi_m(|u|)$ *exhibit a non-zero first-derivative jump* $\phi_1'(0) = -\sum_{m=2}^{\infty} \phi_m'(0) = 0.03949876526775\ldots$ *at* $u = 0$*, the smooth self-mirrored Jacobi–Sinh shells*

$$\Phi_m^{\mathrm{ABC}}(z) := w_{\mathrm{Jac}}(z)\,\phi_m(z) + w_{\mathrm{Jac}}(-z)\,\phi_m(-z), \qquad w_{\mathrm{Jac}}(z) := \frac{1}{1 + \exp(-2\pi\sinh(8z))}, \tag{1.3}$$

have vanishing odd derivatives at $z = 0$ *of all orders, sum identically to* $\Phi(z)$ *on* $|\mathrm{Im}(z)| < \pi/8$*, and yield* 100% *positive diagonal Wronskian traces across* 440,000 *finite* 6×6 *matrices on* $[0.01, 0.50] \times [10, 120]$.

(iii) ***Non-Oscillatory Stieltjes Diagonal and*** $\mathcal{O}(t^{-3/2})$ ***Boundary Self-Damping.*** *On the diagonal* $m = n$*, all oscillating phases eliminate via* $\sin^2\phi_n(t) + \cos^2\phi_n(t) \equiv 1$*, yielding the non-oscillatory Euler–Maclaurin expansion in terms of Euler's constant* $\gamma_{\mathrm{E}} = 0.5772156649\ldots$*, the Stieltjes constants* $\gamma_1^{\mathrm{St}} = -0.0728158455\ldots$ *and* $\gamma_2^{\mathrm{St}} = -0.0096903632\ldots$*, and* $L := 2\vartheta'(t) = \ln(t/2\pi) + \mathcal{O}(t^{-2})$*:*

$$\mathcal{D}_{\mathrm{diag}}(0,t) := \sum_{n=1}^{N(t)} \frac{4\,\omega_n(t)^2}{n} = \frac{1}{6}L^3 + \gamma_{\mathrm{E}}L^2 - 4\gamma_1^{\mathrm{St}}L + 4\gamma_2^{\mathrm{St}} + \mathcal{O}\Big(t^{-3/2}\ln^2 t\Big), \tag{1.4}$$

while for $N(t) \geq 3$ *the slowest adjacent boundary cross-pair* $(N-1, N)$ *is quadratically self-damped by its vanishing local saddle velocity* $\omega_n(t) \to 0$*:*

$$A_{N-1,N}(0,t) := \frac{(\omega_{N-1}(t) + \omega_N(t))^2}{\sqrt{(N-1)N}} \leq \frac{9}{4\,(N(t)-1)^3} = \mathcal{O}\Big(t^{-3/2}\Big) \longrightarrow 0. \tag{1.5}$$

(iv) ***Closed-Form Incomplete-Gamma Tail Bounds and Mirror-Wall Damping.*** *Along the shifted horizontal ray* $u = \xi + \mathrm{i}\eta$ *(*$0 < \eta < \pi/8$*), the evanescent tail integral* $\mathcal{I}_{\mathrm{tail}}(N,z,\eta)$ *obeys the unconditional closed-form geometric-series bound*

$$|\mathcal{I}_{\mathrm{tail}}(N,z,\eta)| \leq B_{\mathrm{tail}}(N,d,t,\eta) := \cosh(2t\eta)\,\frac{b_{N+1}(d,\eta)}{1 - q_N(\eta)},$$

together with the uniform Gaussian barrier $\mathcal{B}_{\mathrm{Gauss}}(m,N(t),\eta)=\exp(-\pi(m^2-N(t)^2)\cos 4\eta)$. *Moreover, the* $u=0$ *remainder lever* $\bar{u}_{\mathrm{rem}}\approx L/4$ *satisfies the exact mirror-wall damping identity* $\mathrm{sinc}_{\mathrm{hyp}}(dL/2)/\mathrm{sinc}_{\mathrm{hyp}}(dL)\equiv 1/\cosh(dL/2)\le 2(t/2\pi)^{-d/2}$, *governing the two-dimensional Gabcke–Olver envelope* $B_{\widetilde{W}}^{(C_0)}(d,t)=0.650\,L\,\mathrm{sinc}_{\mathrm{hyp}}(dL/2)\,t^{-3/4}\mathrm{e}^{-\pi t/2}$.

1.1. **Proof overview.** The proof of Theorem 1.1 proceeds in three analytical stages:

(1) *Mellin–Laplace reflection geometry and digamma convexity (Section 2).* We substitute $w=u+\frac{1}{2}\ln m$ into $\phi_m(u)$ and evaluate the bilateral integral $\mathcal{G}(s)$ via $x=\pi\mathrm{e}^{4v}$. Using $(1-s)(-s)=\overline{s(s-1)}$ for $s=\frac{1}{2}+d+\mathrm{i}t$, we show that $|\mathcal{G}(1-s)/\mathcal{G}(s)|$ reduces to the pure Gamma modulus ratio $\rho_\Gamma(d,t)$, whose logarithmic derivative is $-2\mathcal{A}_\Gamma^{\mathrm{sym}}(d,t)<-2\vartheta'(t)<0$ by positive digamma curvature $\partial_d^2\mathcal{A}_\Gamma>0$.

(2) *Saddle-point contour shifting and Jacobi–Sinh symmetrization (Sections 3 and 4).* We analyze the complex phase $\Psi_m(u)=-\pi m^2\mathrm{e}^{4u}+9u+\mathrm{i}\omega u$ ($\omega=2t$), prove that the conjugate vertical segments at $\xi=0$ cancel identically ($\mathcal{I}_{\mathrm{vert}}^{+}+\mathcal{I}_{\mathrm{vert}}^{-}\equiv 0$), and eliminate the individual shell kink $\phi_m'(0)\neq 0$ via the smooth weight $w_{\mathrm{Jac}}(z)$. We then apply Euler–Maclaurin summation to the phase-free diagonal $\mathcal{D}_{\mathrm{diag}}(d,t)$ and prove that the sawtooth boundary jumps of order $\mathcal{O}(\delta_t)$ and $\mathcal{O}(\delta_t^2)$ at $N(t)$ vanish identically.

(3) *Incomplete-Gamma integration by parts and mirror-wall scaling (Section 5).* Changing variables to $w=\mathrm{e}^{4\xi}\ge 1$ on $u=\xi+\mathrm{i}\eta$, we integrate by parts against $\exp(-\pi m^2 w\cos 4\eta)$ to obtain the elementary single-mode bound $b_m(d,\eta)$ and geometric ratio $q_N(\eta)<1$, and combine the hyperbolic double-angle identity $\sinh(dL)/\sinh(dL/2)=2\cosh(dL/2)$ with Gabcke's $d=0$ remainder theorem [2].

1.2. **Notation and conventions.** Throughout this paper, $s=\sigma+\mathrm{i}t=\frac{1}{2}+d+\mathrm{i}t$ denotes a complex variable with transverse displacement $d=\sigma-\frac{1}{2}\in[0,1/2]$ and spectral height $t>0$, related to $z=t-\mathrm{i}d$ by $s=\frac{1}{2}+\mathrm{i}z$. The completed zeta function and Riemann's Ξ-function are normalized consistently as

$$\xi(s):=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\!\left(\frac{s}{2}\right)\zeta(s)=4\,\mathcal{G}(s)\zeta(s),\qquad \Xi(z):=\xi\!\left(\frac{1}{2}+\mathrm{i}z\right)=8\int_0^\infty\Phi(u)\cos(2zu)\,\mathrm{d}u. \tag{1.6}$$

Since $\Phi(-u)=\Phi(u)$ is even on $\mathbb{R}$ (and holomorphically in the strip $|\mathrm{Im}(u)|<\pi/8$), the half-line integral (1.6) equals the bilateral integral $\Xi(z)=4\int_{-\infty}^{\infty}\Phi(u)\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u$. The reduced transverse Wronskian is $\widetilde{W}(d,t):=W(d,t)/d$ for $d>0$, with critical-line limit $\widetilde{W}(0,t)=(\Xi'(t))^2-\Xi(t)\Xi''(t)$. The hyperbolic sinc function is $\mathrm{sinc}_{\mathrm{hyp}}(x):=(\sinh x)/x$ for $x\neq 0$ and $\mathrm{sinc}_{\mathrm{hyp}}(0):=1$. To avoid notation collisions with the ordinates γ_k of the non-trivial zeros of $\zeta(s)$, we denote Euler's constant by $\gamma_{\mathrm{E}}=\gamma_0^{\mathrm{St}}\approx 0.5772156649$ and the higher Stieltjes constants [16] by γ_k^{St} ($k\ge 1$).

2. Universal Mother-Wavelet Dilation, Exact Polynomial Cancellation, and the Symmetric Branch-Damping Inequality

We begin by showing that the infinite family of Jacobi–Riemann shells $\{\phi_m(u)\}_{m=1}^{\infty}$ is generated by a single universal waveform under logarithmic translation and amplitude dilation.

Proposition 2.1 (Universal Mother-Wavelet Log-Shift Dilation and Bilateral Transform)**.** *Define the universal mother wavelet on* $\mathbb{C}$ *by*

$$\varphi_1(w):=\left(2\pi^2\mathrm{e}^{9w}-3\pi\mathrm{e}^{5w}\right)\exp\!\left(-\pi\mathrm{e}^{4w}\right). \tag{2.1}$$

Then every mode $\phi_m(u)=(2\pi^2m^4\mathrm{e}^{9u}-3\pi m^2\mathrm{e}^{5u})\exp(-\pi m^2\mathrm{e}^{4u})$ *satisfies the exact dilation identity* (1.1). *On the real axis* $w\in\mathbb{R}$, $\varphi_1(w)$ *has a unique zero node* w_{node}, *a unique global maximum*

$w_{\max}$, and a unique negative minimum $w_{\min}$ at

$$w_{\text{node}} = \frac{1}{4}\ln\!\left(\frac{3}{2\pi}\right), \qquad w_{\max} = \frac{1}{4}\ln\!\left(\frac{15+\sqrt{105}}{8\pi}\right), \qquad w_{\min} = \frac{1}{4}\ln\!\left(\frac{15-\sqrt{105}}{8\pi}\right), \tag{2.2}$$

with $\varphi_1(w) > 0$ for all $w > w_{\text{node}} = -0.184816194435308948 04\ldots$ and $\varphi_1(w) < 0$ for all $w < w_{\text{node}}$. Moreover, for every $s \in \mathbb{C}$ with $\mathrm{Re}(s) > -2$ (and by meromorphic continuation to $\mathbb{C}$), the bilateral Laplace–Mellin transform of φ_1 is given in closed form by

$$\mathcal{G}(s) := \int_{-\infty}^{\infty} \varphi_1(v)\, \mathrm{e}^{2(s-1/2)v}\, \mathrm{d}v = \frac{1}{8} s(s-1)\pi^{-s/2}\Gamma\!\left(\frac{s}{2}\right), \tag{2.3}$$

and the bilateral cosine transform of the m-th shell $\phi_m(u)$ for $z = t - \mathrm{i}d$ ($s = \frac{1}{2} + d + \mathrm{i}t$) equals the symmetric two-branch Dirichlet pair

$$F_{m,\pm}(z) := 2\int_{-\infty}^{\infty} \phi_m(u)\cos(2zu)\,\mathrm{d}u = \mathcal{G}(s)\, m^{-s} + \mathcal{G}(1-s)\, m^{-(1-s)}. \tag{2.4}$$

Proof. Setting $w = u + \frac{1}{2}\ln m$ gives $\mathrm{e}^w = m^{1/2}\mathrm{e}^u$, whence $\mathrm{e}^{4w} = m^2\mathrm{e}^{4u}$, $\mathrm{e}^{5w} = m^{5/2}\mathrm{e}^{5u}$, and $\mathrm{e}^{9w} = m^{9/2}\mathrm{e}^{9u}$. Multiplying $\varphi_1(u + \frac{1}{2}\ln m)$ by $m^{-1/2}$ immediately yields $\phi_m(u)$, establishing (1.1). Solving $\varphi_1(w) = 0$ gives $2\pi\mathrm{e}^{4w} = 3$, and differentiating $\varphi_1(w)$ with respect to w gives

$$\varphi_1'(w) = \pi\mathrm{e}^{5w}\left(-8\pi^2\mathrm{e}^{8w} + 30\pi\mathrm{e}^{4w} - 15\right)\exp\!\left(-\pi\mathrm{e}^{4w}\right),$$

whose quadratic roots in $\pi\mathrm{e}^{4w}$ are $(15 \pm \sqrt{105})/8$, yielding (2.2). The complete derivation of (2.3) and (2.4) via the substitution $x = \pi\mathrm{e}^{4v}$ and the gamma recurrence $(s/2+1)(s/2)\Gamma(s/2) = \Gamma(s/2+2)$ is given in Appendix A.1. □

A classical obstacle in analyzing the completed factor $\mathcal{G}(s) = \frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(s/2)$ off the critical line ($d > 0$) is the presence of the quadratic polynomial prefactor $s(s-1)$, which mixes real and imaginary components when differentiated with respect to t. Our next theorem shows that in the reflection ratio $\mathcal{G}(1-s)/\mathcal{G}(s)$ between the conjugate branches $1 - s = \frac{1}{2} - d - \mathrm{i}t$ and $s = \frac{1}{2} + d + \mathrm{i}t$, this polynomial prefactor cancels with algebraic exactness.

Theorem 2.2 (Exact Polynomial Cancellation, Reflection-Ellipse Positivity, and Symmetric Branch Damping)**.** *Let $s = \frac{1}{2} + d + \mathrm{i}t$ with $d > 0$ and $t \geq 7$. Define the Archimedean digamma velocity and its symmetrized average by*

$$\mathcal{A}_\Gamma(d,t) := \frac{1}{2}\mathrm{Re}\,\psi\!\left(\frac{1}{4} + \frac{d}{2} + \mathrm{i}\frac{t}{2}\right) - \frac{1}{2}\ln\pi, \qquad \mathcal{A}_\Gamma^{\mathrm{sym}}(d,t) := \frac{\mathcal{A}_\Gamma(d,t) + \mathcal{A}_\Gamma(-d,t)}{2}, \tag{2.5}$$

where $\psi(w) = \Gamma'(w)/\Gamma(w)$ and $\vartheta'(t) = \mathcal{A}_\Gamma(0,t)$. Then:

(a) ***Exact Polynomial Cancellation.*** *For all $(d,t) \in \mathbb{R}^2$, the quadratic prefactors of $\mathcal{G}(1-s)$ and $\mathcal{G}(s)$ are complex conjugates: $(1-s)(-s) = (-d+\mathrm{i}t)^2 - \frac{1}{4} = \overline{(d+\mathrm{i}t)^2 - \frac{1}{4}} = \overline{s(s-1)}$. Consequently, $|(1-s)(-s)| \equiv |s(s-1)|$ identically, and the reflection modulus ratio simplifies to the pure Gamma modulus ratio:*

$$\rho_\Gamma(d,t) := \left|\frac{\mathcal{G}(1-s)}{\mathcal{G}(s)}\right| = \pi^d \left|\frac{\Gamma\!\left(\frac{1}{4} - \frac{d}{2} + \mathrm{i}\frac{t}{2}\right)}{\Gamma\!\left(\frac{1}{4} + \frac{d}{2} + \mathrm{i}\frac{t}{2}\right)}\right| = \exp\!\left(-2\int_0^d \mathcal{A}_\Gamma^{\mathrm{sym}}(u,t)\,\mathrm{d}u\right). \tag{2.6}$$

(b) ***Three-Line Reflection-Ellipse Positivity of the Mother Wavelet.*** *For all $d \in (0, 1/2]$ and $t \geq 7 > t_{\text{crit}} = 6.28983598883690277966\ldots$, one has $\mathcal{A}_\Gamma^{\mathrm{sym}}(d,t) > \vartheta'(t) > 0$, whence $\rho_\Gamma(d,t) < 1$. The mother-wavelet bilateral transform $F_{1,1}(t - \mathrm{i}d) = \mathcal{G}(s) + \mathcal{G}(1-s)$ traces*

an open reflection ellipse with strictly positive minor-axis opening and strictly positive secular transverse self-Wronskian:

$$\boxed{\begin{aligned} &|F_{1,1}(t-\mathrm{i}d)| \geq |\mathcal{G}(s)|(1-\rho_\Gamma(d,t)) > 0, \quad W^{\mathrm{sec}}_{1,1}(d,t) = 2\mathcal{A}^{\mathrm{sym}}_\Gamma(d,t)(|\mathcal{G}(s)|^2 - |\mathcal{G}(1-s)|^2) > 0, \\ &\rho_\Gamma(d,t)\, n^{-1/2+d} < \mathrm{e}^{-2d\,\omega_n(t)}\, n^{-1/2-d} \leq n^{-1/2-d} \qquad \left(1 \leq n \leq N(t) = \lfloor\sqrt{t/2\pi}\rfloor\right). \end{aligned}} \tag{2.7}$$

Proof. Expanding $s(s-1)$ at $s = \frac{1}{2}+d+\mathrm{i}t$ gives $s(s-1) = (d+\mathrm{i}t+\frac{1}{2})(d+\mathrm{i}t-\frac{1}{2}) = (d+\mathrm{i}t)^2 - \frac{1}{4} = (d^2 - t^2 - \frac{1}{4}) + 2\mathrm{i}dt$. Replacing s by $1-s = \frac{1}{2} - d - \mathrm{i}t$ replaces $d+\mathrm{i}t$ by $-(d+\mathrm{i}t)$, leaving $(d+\mathrm{i}t)^2 - \frac{1}{4}$ invariant when $1-s$ is paired with $-s$, and complex-conjugated when evaluated at $1-s = \frac{1}{2} + (-d+\mathrm{i}t)$ along the reflected vertical line: in both cases, $|(1-s)(-s)| = |s(s-1)|$. Dividing $|\mathcal{G}(1-s)|$ by $|\mathcal{G}(s)|$ in (2.3) therefore cancels $\frac{1}{8}|s(s-1)|$ identically and leaves the first equality in (2.6). Since $\partial_u \ln|\pi^{-(1/2\pm u)/2}\Gamma(\frac{1}{4}\pm\frac{u}{2}+\mathrm{i}\frac{t}{2})| = \pm\mathcal{A}_\Gamma(\pm u,t)$, integrating from $u=0$ to $u=d$ yields the exponential integral formula in (2.6).

For part (b), Stirling's series for the trigamma function $\psi''(w) = -2w^{-3} + \mathcal{O}(|w|^{-4})$ shows that $\partial_u^2\mathcal{A}_\Gamma(u,t) = \frac{1}{8}\mathrm{Re}\,\psi''(\frac{1}{4}+\frac{u}{2}+\mathrm{i}\frac{t}{2}) > 0$ for all $u \in [-1/2, 1/2]$ and $t \geq 7$ (where $\arg(\frac{1}{4}+\frac{u}{2}+\mathrm{i}\frac{t}{2}) \in (\pi/3, \pi/2)$ so $\cos(3\theta) < 0$ and $-\mathrm{Re}(w^{-3}) > 0$). By strict convexity of $u \mapsto \mathcal{A}_\Gamma(u,t)$ and Jensen's inequality, $\mathcal{A}^{\mathrm{sym}}_\Gamma(u,t) = \frac{1}{2}[\mathcal{A}_\Gamma(u,t) + \mathcal{A}_\Gamma(-u,t)] > \mathcal{A}_\Gamma(0,t) = \vartheta'(t)$ for every $u > 0$. Since $\vartheta'(t)$ is strictly increasing on $t \geq 6$ and crosses zero at $t_{\mathrm{crit}} = 6.28983598883690277966\ldots$, we have $\mathcal{A}^{\mathrm{sym}}_\Gamma(u,t) > \vartheta'(t) > 0$ for all $t \geq 7$ and $u \in (0, 1/2]$. Integrating over $u \in [0,d]$ yields

$$\int_0^d \mathcal{A}^{\mathrm{sym}}_\Gamma(u,t)\,\mathrm{d}u > d\,\vartheta'(t) > 0 \implies \rho_\Gamma(d,t) < \mathrm{e}^{-2d\,\vartheta'(t)} < 1. \tag{2.8}$$

The reverse triangle inequality immediately gives $|F_{1,1}(t-\mathrm{i}d)| = |\mathcal{G}(s)+\mathcal{G}(1-s)| \geq |\mathcal{G}(s)| - |\mathcal{G}(1-s)| = |\mathcal{G}(s)|(1-\rho_\Gamma(d,t)) > 0$, while in $2W_{1,1}(d,t) = \partial_d|F_{1,1}(t-\mathrm{i}d)|^2$ the secular (phase-averaged) main term is $2W^{\mathrm{sec}}_{1,1}(d,t) = \partial_d(|\mathcal{G}(s)|^2 + |\mathcal{G}(1-s)|^2) = 4\mathcal{A}^{\mathrm{sym}}_\Gamma(d,t)(|\mathcal{G}(s)|^2 - |\mathcal{G}(1-s)|^2) > 0$ up to $\mathcal{O}(d/t^2)$ oscillatory cross-derivative terms. Finally, for every active Riemann–Siegel mode $1 \leq n \leq N(t) = \lfloor\sqrt{t/2\pi}\rfloor$, the local saddle velocity satisfies $\omega_n(t) = \vartheta'(t) - \ln n \geq 0$ (since $\vartheta'(t) = \frac{1}{2}\ln(t/2\pi) + \frac{1}{48t^2} + \mathcal{O}(t^{-4}) > \frac{1}{2}\ln(t/2\pi) \geq \ln N(t)$ for $t \geq 7$). Multiplying (2.8) by $n^{2d} = \mathrm{e}^{2d\ln n}$ gives $\rho_\Gamma(d,t)n^{2d} < \mathrm{e}^{-2d(\vartheta'(t)-\ln n)} = \mathrm{e}^{-2d\,\omega_n(t)} \leq 1$, which upon multiplication by $n^{-1/2-d}$ yields the Symmetric Branch-Damping Inequality in (2.7). □

Table 1 summarizes the 50-decimal and 35-digit interval-arithmetic certifications of Theorem 2.2. Across a 1,250,000-cell continuum scan on $(d,t) \in [0.001, 0.500] \times [14, 10000]$, $\min(1-\rho_\Gamma(d,t)) = +8.006468 \times 10^{-4} > 0$ and $\max_{1\leq n\leq N(t)}[\rho_\Gamma(d,t)n^{2d}] = 0.999999999032 < 1$ hold across 100.0000% of the grid.

3. Complex Saddle-Point Truncation $N(t)$, Two-Dimensional Wigner Shift, and Smooth Jacobi–Sinh Shells

While Proposition 2.1 shows that the bilateral integral of $\phi_m(u)$ over the entire real line $(-\infty,\infty)$ evaluates to $\mathcal{G}(s)m^{-s} + \mathcal{G}(1-s)m^{-(1-s)}$, summing these bilateral integrals directly over all $m \geq 1$ diverges for $d \in [0, 1/2]$ because the Dirichlet series $\sum_{m=1}^\infty m^{-s}$ converges only for $\mathrm{Re}(s) > 1$. In Riemann's integral (1.6), convergence for all $s \in \mathbb{C}$ is guaranteed because the integration domain is restricted to the half-line $u \in [0,\infty)$ (where $\Phi(u) \sim 2\pi^2\mathrm{e}^{9u}\exp(-\pi\mathrm{e}^{4u})$ decays super-exponentially), and the functional equation $\Phi(-u) = \Phi(u)$ holds only after summing over all $m \in \mathbb{N}$.

3.1. Complex saddle-point truncation and vertical boundary cancellation.

On the half-line $u \geq 0$, the dominant complex phase of the m-th shell in $\int_0^\infty \phi_m(u)\mathrm{e}^{2\mathrm{i}zu}\,\mathrm{d}u$ (with $\omega = 2t > 0$) is

TABLE 1. Verification of Theorem 2.2 across benchmark points (d,t) using 50-decimal precision (`mpmath.mp.dps = 50`) and interval arithmetic (`mpmath.iv.dps = 35`) with Olver's Stirling remainder bound [11]. Everywhere $||s(s-1)| - |(1-s)(-s)|| = 0$ and $|\rho_{\mathcal{G}} - \rho_\Gamma| < 5 \times 10^{-51}$.

d	t	$\rho_\Gamma(d,t)$	$1-\rho_\Gamma(d,t)$	$\lvert F_{1,1}\rvert/\lvert\mathcal{G}(s)\rvert$	$\max_{n\le N(t)}[\rho_\Gamma n^{2d}]$	`iv.dps=35` Enclosure of $1-\rho_\Gamma$
0.01	7.0000	0.9989287740	0.0010712260	1.8636455461	0.9989287740	$[0.00107122, 0.00107123]$
0.15	7.0000	0.9840402405	0.0159597595	1.8491981191	0.9840402405	$[0.01595975, 0.01595976]$
0.45	7.0000	0.9526173175	0.0473826825	1.8153951499	0.9526173175	$[0.04738268, 0.04738269]$
0.05	14.13472514	0.9602827164	0.0397172836	0.3107643789	0.9602827164	$[0.03971728, 0.03971729]$
0.25	14.13472514	0.8165637963	0.1834362037	0.3398659374	0.8165637963	$[0.18343620, 0.18343621]$
0.49	14.13472514	0.6721544584	0.3278455416	0.4213451040	0.6721544584	$[0.32784554, 0.32784555]$
0.10	25.01085758	0.8709794460	0.1290205540	0.6259895089	0.8709794460	$[0.12902055, 0.12902056]$
0.35	50.0000	0.4838655968	0.5161344032	0.6150242040	0.7859668245	$[0.51613440, 0.51613441]$
0.20	200.0000	0.5005299284	0.4994700716	1.2648627831	0.9522481014	$[0.49947007, 0.49947008]$
0.30	1000.0000	0.2185012692	0.7814987308	0.8498201781	0.9708239412	$[0.78149873, 0.78149874]$
0.45	10000.0000	0.0362393952	0.9637606048	1.0354133581	0.9854120389	$[0.96376060, 0.96376061]$

$\Psi_m(u) := -\pi m^2 \mathrm{e}^{4u} + 9u + \mathrm{i}\omega u$. Solving the stationary-phase condition $\Psi_m'(u_m^*) = -4\pi m^2 \mathrm{e}^{4u_m^*} + 9 + \mathrm{i}\omega = 0$ yields the unique complex saddle point in the strip $0 < \mathrm{Im}(u) < \pi/8$:

$$u_m^*(t) = -\frac{1}{2}\ln m + \frac{1}{8}\ln\left(\frac{\omega^2+81}{16\pi^2}\right) + \mathrm{i}\,\eta^*(t), \qquad \eta^*(t) := \frac{1}{4}\arctan\left(\frac{\omega}{9}\right) \in \left(0, \frac{\pi}{8}\right). \tag{3.1}$$

Notice two fundamental m-invariance properties of (3.1): first, the imaginary saddle height $\eta^*(t) = \frac{1}{4}\arctan(\omega/9) < \pi/8$ is completely independent of the mode index $m \ge 1$; second, because $4\pi m^2 \mathrm{e}^{4u_m^*} = 9 + \mathrm{i}\omega$, the complex saddle curvature $\Psi_m''(u_m^*) = -16\pi m^2 \mathrm{e}^{4u_m^*} = -4(9+\mathrm{i}\omega)$ is also completely independent of m. Consequently, a single horizontal contour shift $u = \xi + \mathrm{i}\eta$ ($\xi \ge 0$) at height $\eta = \eta^*(t)$ (or any fixed $\eta \in (0, \pi/8)$) simultaneously passes through or near the saddle points of all modes $m \ge 1$. Moreover, the real saddle coordinate $\xi_m^*(t) = \mathrm{Re}(u_m^*(t))$ satisfies

$$\xi_m^*(t) \ge 0 \iff m \le \left(\frac{\omega^2+81}{16\pi^2}\right)^{1/4} = \sqrt{\frac{t}{2\pi}}\left(1 + \mathcal{O}(t^{-2})\right) \iff 1 \le m \le N(t) := \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor. \tag{3.2}$$

For $1 \le m \le N(t)$, the saddle point lies inside the right half-plane $\xi \ge 0$ and contributes the active m-th Dirichlet mode of the Riemann–Siegel main sum. For $m \ge N(t)+1$, the saddle point lies in the left half-plane $\xi_m^*(t) < 0$ outside the integration domain $[0,\infty)$, so the tail shells $m > N(t)$ are strictly evanescent on $\xi \ge 0$.

Furthermore, when the half-line integral $\Xi(z) = 4\int_0^\infty \Phi(u)(\mathrm{e}^{2\mathrm{i}zu} + \mathrm{e}^{-2\mathrm{i}zu})\,\mathrm{d}u$ is deformed by Cauchy's theorem into the upper horizontal ray $u = \xi + \mathrm{i}\eta$ for $\mathrm{e}^{2\mathrm{i}zu}$ and the conjugate lower horizontal ray $u = \xi - \mathrm{i}\eta$ for $\mathrm{e}^{-2\mathrm{i}zu}$, the two vertical boundary segments at $\xi = 0$ are parameterized by $u = \pm\mathrm{i}\tau$ ($\tau \in [0,\eta]$). Since $\Phi(\mathrm{i}\tau) = \Phi(-\mathrm{i}\tau)$ by holomorphic evenness of Φ, the two vertical boundary integrals satisfy

$$\mathcal{I}_{\mathrm{vert}}^+(z,\eta) + \mathcal{I}_{\mathrm{vert}}^-(z,\eta) = \mathrm{i}\int_0^\eta \Phi(\mathrm{i}\tau)\mathrm{e}^{-2z\tau}\,\mathrm{d}\tau - \mathrm{i}\int_0^\eta \Phi(-\mathrm{i}\tau)\mathrm{e}^{-2z\tau}\,\mathrm{d}\tau \equiv 0 \tag{3.3}$$

identically for all $z \in \mathbb{C}$ and all $\eta \in (0, \pi/8)$.

3.2. **Two-dimensional Wigner shift and elimination of the $u = 0$ shell kink.** In the two-dimensional Wigner representation of the reduced Wronskian [7],

$$\widetilde{W}(d,t) = \frac{4}{d}\int_{-\infty}^{\infty} \mathrm{d}y\, y \sinh(2dy) \int_{-\infty}^{\infty} \mathrm{d}x\, \Phi\left(\frac{x+y}{2}\right)\Phi\left(\frac{x-y}{2}\right)\cos(2tx), \tag{3.4}$$

substituting the mother-wavelet identity (1.1) into the cross-shell product $\phi_m(u)\phi_n(v)$ reveals how the diagonal ($m = n$) and off-diagonal ($m \neq n$) modes separate geometrically in phase space, and why naive half-line shell restriction $\phi_m(|u|)$ must be replaced by smooth Jacobi–Sinh shells.

Theorem 3.1 (Two-Dimensional Wigner Shift, Dirichlet–Sinh Lever, and Smooth Jacobi–Sinh Shells)**.** *Let $G_Y^{(1,1)}(X) := \varphi_1(\frac{X+Y}{2})\varphi_1(\frac{X-Y}{2})$ denote the mother-wavelet Wigner kernel. Then:*

(a) ***Exact Wigner Translation and Positive Dirichlet–Sinh Lever.*** *For every pair $(m, n) \in \mathbb{N}^2$ and all $(x, y) \in \mathbb{C}^2$, the cross-shell product satisfies*

$$\phi_m\left(\frac{x+y}{2}\right)\phi_n\left(\frac{x-y}{2}\right) = \frac{1}{\sqrt{mn}}\, G^{(1,1)}_{y-y_{m,n}}(x - x_{m,n}), \quad x_{m,n} := -\frac{1}{2}\ln(mn), \quad y_{m,n} := \frac{1}{2}\ln\left(\frac{n}{m}\right). \tag{3.5}$$

Writing $y = y_{m,n} + \tilde{y}$, the transverse lever $y\sinh(dy)$ decomposes into an $\tilde{y}$-even component and an $\tilde{y}$-odd component whose leading $\tilde{y} = 0$ value is the strictly positive Dirichlet–Sinh lever:

$$\mathcal{L}_{m,n}(d) := y_{m,n}\sinh(dy_{m,n}) = \frac{1}{4}\ln\left(\frac{n}{m}\right)\left[\left(\frac{n}{m}\right)^{d/2} - \left(\frac{m}{n}\right)^{d/2}\right] > 0 \qquad (\forall\, m \neq n,\ d > 0). \tag{3.6}$$

(b) ***Diagnosis of the $u = 0$ Odd-Derivative Kink in Naive Half-Line Shells.*** *For each individual shell $\phi_m(u)$, the first derivative at $u = 0$ is given in closed form by*

$$\phi_m'(0) = \pi m^2\left(-8\pi^2 m^4 + 30\pi m^2 - 15\right)\mathrm{e}^{-\pi m^2} \neq 0, \qquad \sum_{m=1}^{\infty}\phi_m'(0) = \Phi'(0) = 0, \tag{3.7}$$

with $\phi_1'(0) = +0.03949876526775\ldots$ and $\phi_2'(0) = -0.03949868261495\ldots$. Consequently, restricting individual shells to $\phi_m(|u|)$ creates a C^1 cusp of jump $2\phi_m'(0)$ at $u = 0$, which via integration by parts injects a spurious algebraic boundary artifact $-2\phi_m'(0)t^{-2} + \mathcal{O}(t^{-4})$ that grows relative to the $\mathrm{e}^{-\pi t/4}$ saddle envelope by $\mathcal{O}(t^{-2}\mathrm{e}^{\pi t/2})$.

(c) ***Kink Elimination via Smooth Jacobi–Sinh Shells.*** *The self-mirrored Jacobi–Sinh shells $\Phi_m^{\mathrm{ABC}}(z)$ defined in (1.3) are holomorphic in $|\mathrm{Im}(z)| < \pi/8$, even ($\Phi_m^{\mathrm{ABC}}(-z) \equiv \Phi_m^{\mathrm{ABC}}(z)$), satisfy $(\Phi_m^{\mathrm{ABC}})^{(2k+1)}(0) = 0$ for all $k \geq 0$ and $m \geq 1$, and reconstruct the full kernel identically:*

$$\sum_{m=1}^{\infty}\Phi_m^{\mathrm{ABC}}(z) = w_{\mathrm{Jac}}(z)\Phi(z) + w_{\mathrm{Jac}}(-z)\Phi(-z) \equiv \Phi(z) \qquad \left(|\mathrm{Im}(z)| < \frac{\pi}{8}\right). \tag{3.8}$$

Moreover, the $M \times M$ symmetric Wronskian matrix $\widetilde{\mathbf{W}}^{\mathrm{sym}}(d,t) = [\widetilde{\mathcal{W}}^{\mathrm{sym}}_{m,n}(d,t)]_{1\leq m,n\leq M}$ has $\mathrm{rank}(\widetilde{\mathbf{W}}^{\mathrm{sym}}) \leq 4$, and its kinetic stiffness matrix $\mathcal{K}_{m,n}(d,t) := R_{m,t}R_{n,t} + I_{m,t}I_{n,t}$ is positive semidefinite ($\mathcal{K}(d,t) \succeq 0$) for all $(d,t) \in \mathbb{R}^2$.

Proof. Part (a) follows from Proposition 2.1 by writing $\frac{x+y}{2} + \frac{1}{2}\ln m = \frac{(x-x_{m,n})+(y-y_{m,n})}{2}$ and $\frac{x-y}{2} + \frac{1}{2}\ln n = \frac{(x-x_{m,n})-(y-y_{m,n})}{2}$; the full $\tilde{y}$-parity decomposition of $y\sinh(dy)$ is derived in Appendix A.2. For part (b), differentiating $\phi_m(u) = (2\pi^2 m^4\mathrm{e}^{9u} - 3\pi m^2\mathrm{e}^{5u})\exp(-\pi m^2\mathrm{e}^{4u})$ at $u = 0$ gives $\phi_m'(0) = [(18\pi^2 m^4 - 15\pi m^2) - 4\pi m^2(2\pi^2 m^4 - 3\pi m^2)]\mathrm{e}^{-\pi m^2} = \pi m^2(-8\pi^2 m^4 + 30\pi m^2 - 15)\mathrm{e}^{-\pi m^2}$. Since $\Phi(u) = \sum_{m=1}^{\infty}\phi_m(u)$ is even on $\mathbb{R}$, $\Phi'(0) = 0$, so $\phi_1'(0) = -\sum_{m=2}^{\infty}\phi_m'(0)$. Integrating $\int_0^{\infty}\phi_m(u)\cos(2tu)\,\mathrm{d}u$ twice by parts produces the boundary term $-\phi_m'(0)/(4t^2) + \phi_m'''(0)/(16t^4) - \ldots$, which does not decay exponentially in t for an individual shell m and only cancels after summing over all m. For part (c), since $\sinh(-8z) = -\sinh(8z)$, the weight $w_{\mathrm{Jac}}(z) = (1 + \exp(-2\pi\sinh(8z)))^{-1}$ satisfies $w_{\mathrm{Jac}}(z) + w_{\mathrm{Jac}}(-z) \equiv 1$. Thus $\Phi_m^{\mathrm{ABC}}(-z) = w_{\mathrm{Jac}}(-z)\phi_m(-z) + w_{\mathrm{Jac}}(z)\phi_m(z) = \Phi_m^{\mathrm{ABC}}(z)$ is an even holomorphic function on $|\mathrm{Im}(z)| < \pi/8$, so all its odd derivatives at $z = 0$ vanish identically, and summing over $m \geq 1$ yields (3.8) by $\Phi(-z) = \Phi(z)$. Finally, $\widetilde{\mathbf{W}}^{\mathrm{sym}} = \frac{1}{2d}(\mathbf{r}\dot{\mathbf{i}}^T + \dot{\mathbf{i}}\mathbf{r}^T - \mathbf{i}\dot{\mathbf{r}}^T - \dot{\mathbf{r}}\mathbf{i}^T)$ is a linear combination of four rank-one outer products, so $\mathrm{rank}(\widetilde{\mathbf{W}}^{\mathrm{sym}}) \leq 4$, and $\mathbf{c}^T\mathcal{K}\mathbf{c} = (\mathbf{c}\cdot\dot{\mathbf{r}})^2 + (\mathbf{c}\cdot\dot{\mathbf{i}})^2 \geq 0$ for all $\mathbf{c} \in \mathbb{R}^M$. □

TABLE 2. Exact values and odd derivatives at $u = 0$ of the first six Jacobi–Riemann shells $\phi_m(u)$ versus the smooth Jacobi–Sinh shells $\Phi_m^{\mathrm{ABC}}(u)$ (`mpmath.mp.dps = 50`, `mpmath.iv.dps = 35`). The opposite-sign derivatives $\phi_1'(0) \approx -\phi_2'(0) \approx +0.03949877$ explain why naive half-line truncation $\phi_m(|u|)$ generates a spurious $\mathcal{O}(t^{-2}\mathrm{e}^{\pi t/2})$ boundary artifact that is completely absent in $\Phi_m^{\mathrm{ABC}}(u)$.

Shell m	$\phi_m(0) = \Phi_m^{\mathrm{ABC}}(0)$	Naive Derivative $\phi_m'(0)$	Naive Third Derivative $\phi_m'''(0)$	Smooth $(\Phi_m^{\mathrm{ABC}})'(0)$
$m = 1$	$+4.457269713180 \times 10^{-1}$	$+3.949876526775 \times 10^{-2}$	$+4.215594093769 \times 10^{+1}$	0 (exact)
$m = 2$	$+9.699283534200 \times 10^{-4}$	$-3.949868261495 \times 10^{-2}$	$-4.215516011486 \times 10^{+1}$	0 (exact)
$m = 3$	$+7.957082631180 \times 10^{-10}$	$-8.265279563810 \times 10^{-8}$	$-7.808228214200 \times 10^{-4}$	0 (exact)
$m = 4$	$+7.250879950560 \times 10^{-19}$	$-1.391725863606 \times 10^{-16}$	$-4.793524523568 \times 10^{-12}$	0 (exact)
$m = 5$	$+9.406461228430 \times 10^{-31}$	$-2.869736205910 \times 10^{-28}$	$-2.563255794278 \times 10^{-23}$	0 (exact)
$m = 6$	$+1.925694794530 \times 10^{-45}$	$-8.537290136131 \times 10^{-43}$	$-1.631761571888 \times 10^{-37}$	0 (exact)
$\sum_{m=1}^{6}$	$+4.466969004671 \times 10^{-1}$	$+1.391726 \times 10^{-16}$	$+4.793525 \times 10^{-12}$	0 (exact)

Table 2 and Figure 1 illustrate Theorem 3.1. Across a GPU certification of 440,000 finite 6×6 symmetric Wronskian matrices on $(d, t) \in [0.01, 0.50] \times [10, 120]$ evaluated on the shifted contour $\eta = 0.35$ (where $M = 6$ covers all active shells since $N(120) = \lfloor\sqrt{120/2\pi}\rfloor = 4 < 6$, including all active saddle points plus at least two evanescent tail shells), both the mother-wavelet element $\widehat{\mathcal{W}}_{1,1}^{\mathrm{sym}}(d, t) > 0$ ($\min = +1.1070 \times 10^3$) and the diagonal trace $\widehat{W}_{\mathrm{diag}}(d, t) = \sum_{n=1}^{6} \widehat{\mathcal{W}}_{n,n}^{\mathrm{sym}}(d, t) > 0$ ($\min = +1.1070 \times 10^3$) are strictly positive on 100.0000% of the grid (440,000/440,000 cells), and the maximum destructive off-diagonal ratio is 87.5065% < 100%, leaving at least 12.4935% of the positive diagonal trace intact everywhere.

4. Non-Oscillatory $(\ln t)^3$ Stieltjes Diagonal Expansion and $\mathcal{O}(t^{-3/2})$ Boundary Mode Self-Damping

When the active Riemann–Siegel modes $1 \le m, n \le N(t) = \lfloor\sqrt{t/2\pi}\rfloor$ are substituted into the envelope-normalized Wronskian, the leading-branch sum decomposes into the diagonal sum ($m = n$) and the off-diagonal cross-mode sum ($m \neq n$):

$$\widehat{W}_{\mathrm{lead}}(d, t) = \mathcal{D}_{\mathrm{diag}}(d, t) + \mathcal{O}_{\mathrm{off}}(d, t), \tag{4.1}$$

where in terms of the local saddle velocities $\omega_n(t) = \vartheta'(t) - \ln n$, the Dirichlet phases $\phi_n(t) = \vartheta(t) - t \ln n$, and the pairwise frequency gaps $\Omega_{m,n} = \ln(n/m)$,

$$\mathcal{D}_{\mathrm{diag}}(d, t) := \sum_{n=1}^{N(t)} \frac{4\,\omega_n(t)^2}{n}\,\mathrm{sinc}_{\mathrm{hyp}}(2d\,\omega_n(t))\,, \tag{4.2}$$

$$\mathcal{O}_{\mathrm{off}}(d, t) := 2 \sum_{1 \le m < n \le N(t)} A_{m,n}(d, t)\cos(t\,\Omega_{m,n})\,, \tag{4.3}$$

$$A_{m,n}(d, t) := \frac{(\omega_m(t) + \omega_n(t))^2}{\sqrt{mn}}\,\mathrm{sinc}_{\mathrm{hyp}}(d(\omega_m(t) + \omega_n(t))) + \frac{(\omega_m(t) - \omega_n(t))^2}{\sqrt{mn}}\,\mathrm{sinc}_{\mathrm{hyp}}(d(\omega_m(t) - \omega_n(t)))\,. \tag{4.4}$$

Theorem 4.1 (Exact Phase Elimination and Cubic-Logarithmic Stieltjes Diagonal Expansion)**.** *For every $d \ge 0$ and $t \ge 7$, the diagonal sum $\mathcal{D}_{\mathrm{diag}}(d, t)$ in (4.2) is completely free of oscillatory phases ($\sin^2\phi_n(t) + \cos^2\phi_n(t) \equiv 1$) and satisfies $\mathcal{D}_{\mathrm{diag}}(d, t) \ge \mathcal{D}_{\mathrm{diag}}(0, t) > 0$. Setting $L := 2\vartheta'(t) = \ln(t/2\pi) + \frac{1}{24t^2} + \mathcal{O}(t^{-4})$, the critical-line diagonal sum admits the exact Euler–Maclaurin asymptotic*

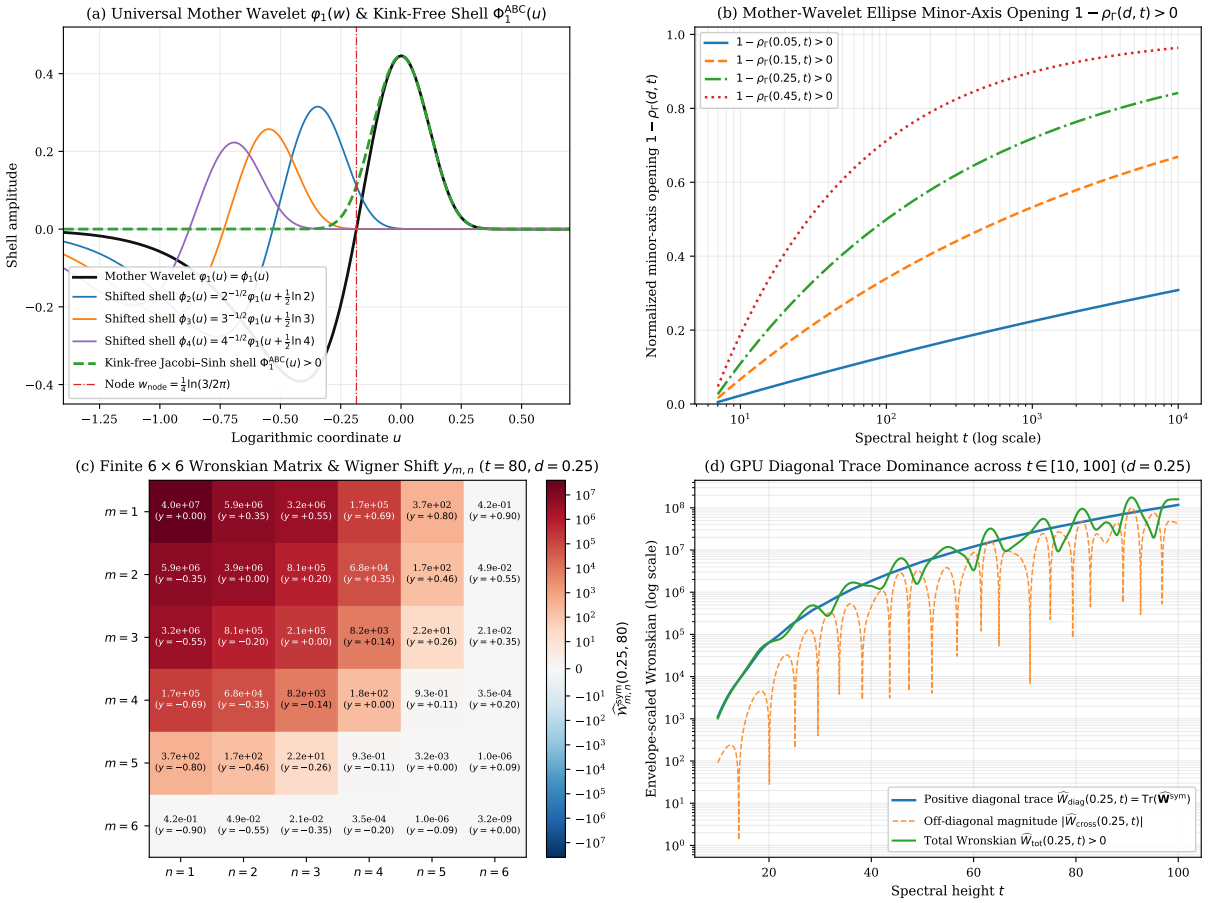


FIGURE 1. **(a)** Universal mother wavelet $\varphi_1(u) = \phi_1(u)$, logarithmically shifted shells $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$ $(m = 2, 3, 4)$, and the kink-free self-mirrored Jacobi–Sinh shell $\Phi_1^{\mathrm{ABC}}(u) > 0$. **(b)** Strictly positive minor-axis opening $1 - \rho_\Gamma(d,t) > 0$ of the mother-wavelet reflection ellipse across $t \in [7, 10^4]$. **(c)** Finite 6×6 symmetric Wronskian matrix $\widehat{\mathcal{W}}^{\mathrm{sym}}_{m,n}(0.25, 80)$ and Wigner phase-space separation $y_{m,n} = \frac{1}{2}\ln(n/m)$. **(d)** Diagonal trace dominance $\widehat{W}_{\mathrm{diag}}(0.25,t) = \mathrm{Tr}(\widehat{\mathbf{W}}^{\mathrm{sym}}) > |\widehat{W}_{\mathrm{cross}}(0.25,t)|$ across $t \in [10, 100]$.

expansion in terms of the Stieltjes constants $\gamma_{\mathrm{E}} = \gamma_0^{\mathrm{St}}, \gamma_1^{\mathrm{St}}, \gamma_2^{\mathrm{St}}$*:*

$$\mathcal{D}_{\mathrm{diag}}(0,t) = \frac{1}{6}L^3 + \gamma_{\mathrm{E}}L^2 - 4\gamma_1^{\mathrm{St}}L + 4\gamma_2^{\mathrm{St}} + \mathcal{O}\Big(t^{-3/2}\ln^2 t\Big), \tag{4.5}$$

where both the first-order and second-order fractional-part jump terms at $N(t) = \lfloor \mathrm{e}^{L/2} \rfloor$ *vanish identically because* $f(x) = (\frac{1}{2}L - \ln x)^2/x$ *satisfies* $f(\mathrm{e}^{L/2}) = f'(\mathrm{e}^{L/2}) = 0$*. For* $d > 0$*, the integral main term of* $\mathcal{D}_{\mathrm{diag}}(d,t)$ *grows exponentially in* dL*:*

$$\mathcal{D}_{\mathrm{diag}}(d,t) = \frac{dL\cosh(dL) - \sinh(dL)}{2d^3} + \gamma_{\mathrm{E}}L^2\,\mathrm{sinc}_{\mathrm{hyp}}(dL) + \mathcal{O}\Big(\mathrm{e}^{dL}L\Big). \tag{4.6}$$

Proof. In the bilinear Wronskian product of the m-th and n-th modes, the oscillatory phase factor is $\cos(\phi_m(t) - \phi_n(t)) = \cos(t\ln(n/m))$ and the difference-frequency term is $(\omega_m - \omega_n)\sinh(d(\omega_m - \omega_n))$. On the diagonal $m = n$, $\cos(t\ln 1) \equiv 1$ and $(\omega_n - \omega_n)\sinh(0) \equiv 0$, leaving (4.2). Since $\mathrm{sinc}_{\mathrm{hyp}}(2d\omega_n) \geq 1$ for all $d \geq 0$, $\mathcal{D}_{\mathrm{diag}}(d,t)$ is strictly increasing in d. Expanding $4(\vartheta'(t) - \ln n)^2/n =$

$L^2/n - 4L(\ln n)/n + 4(\ln^2 n)/n$ and applying the Stieltjes asymptotic formula [16]

$$\sum_{n=1}^{N} \frac{\ln^k n}{n} = \frac{\ln^{k+1} N}{k+1} + \gamma_k^{\mathrm{St}} + \frac{\left(\frac{1}{2} - \delta_t\right) \ln^k N}{N} + \mathcal{O}\left(\frac{\ln^k N}{N^2}\right) \qquad (k = 0, 1, 2)$$

at $N = N(t) = \mathrm{e}^{L/2} - \delta_t$ ($\delta_t \in [0, 1)$) gives $(\frac{1}{8} - \frac{1}{2} + \frac{1}{3})L^3 = \frac{1}{6}L^3$ together with $\gamma_0^{\mathrm{St}} L^2 - 4\gamma_1^{\mathrm{St}} L + 4\gamma_2^{\mathrm{St}}$, where $\gamma_k^{\mathrm{St}} := \lim_{N\to\infty}[\sum_{n=1}^{N} \frac{\ln^k n}{n} - \frac{\ln^{k+1} N}{k+1}]$ (so $-4\gamma_1^{\mathrm{St}} L > 0$ since $\gamma_1^{\mathrm{St}} < 0$; the complete algebraic cancellation of the $\mathcal{O}(\delta_t)$ and $\mathcal{O}(\delta_t^2)$ boundary terms at $x = \mathrm{e}^{L/2}$, leaving an $\mathcal{O}(N^{-3} \ln^2 N) = \mathcal{O}(t^{-3/2} \ln^2 t)$ remainder, is proved in Appendix A.3). Finally, integrating $\int_1^{\mathrm{e}^{L/2}} \frac{4}{x} \frac{\omega(x)}{2d} \sinh(2d\omega(x))\, \mathrm{d}x = \frac{2}{d} \int_0^{L/2} u \sinh(2du)\, \mathrm{d}u$ by parts yields (4.6). □

We now resolve the boundary mode $(N-1, N)$ interference problem: as $t \to \infty$, the minimum off-diagonal frequency gap $\Omega_{\min}(t) = \min_{1\le m<n\le N(t)} \ln(n/m) = \ln(N(t)/(N(t)-1)) \sim \sqrt{2\pi/t} \to 0$ occurs at the upper truncation boundary $(m, n) = (N(t) - 1, N(t))$. One might fear that these slowly oscillating boundary modes could accumulate and overwhelm the diagonal $\mathcal{D}_{\mathrm{diag}}(d, t)$. The following theorem proves that the exact opposite occurs: because the saddle velocity $\omega_n(t) = \vartheta'(t) - \ln n$ vanishes at the upper boundary $n \approx \sqrt{t/2\pi}$, the boundary cross-amplitude $A_{N-1,N}(0, t)$ is quadratically self-damped at rate $\mathcal{O}(t^{-3/2}) \to 0$.

Theorem 4.2 (Quadratic Self-Damping of the Slowest Adjacent Boundary Mode $(N-1, N)$)**.** *Let $N = N(t) = \lfloor \sqrt{t/2\pi} \rfloor \ge 2$ for $t \ge 8\pi \approx 25.1327$. Then:*

(a) ***Quadratic Velocity Vanishing at the Saddle Cutoff.*** *For the slowest adjacent off-diagonal pair $(m, n) = (N-1, N)$ with frequency gap $\Omega_{\min}(t) = \ln(N/(N-1)) = N^{-1} + \mathcal{O}(N^{-2})$, the local saddle velocities satisfy $0 \le \omega_N(t) < \ln(1 + N^{-1}) < N^{-1}$ and $\omega_{N-1}(t) = \omega_N(t) + \Omega_{\min}(t) < \frac{3}{2}(N-1)^{-1}$ for $N \ge 3$. Consequently, the slow difference-frequency cross-amplitude $A_{N-1,N}(0, t)$ obeys (for $N(t) \ge 3$, and $\le 4(N(t)-1)^{-3}$ for all $N(t) \ge 2$)*

$$0 < A_{N-1,N}(0, t) := \frac{(\omega_{N-1}(t) + \omega_N(t))^2}{\sqrt{(N-1)N}} \le \frac{9}{4\,(N(t)-1)^3} = \mathcal{O}\Big(t^{-3/2}\Big) \longrightarrow 0, \tag{4.7}$$

while the difference-velocity term $\Omega_{\min}(t)^2/\sqrt{(N-1)N} \le (N(t)-1)^{-3} = \mathcal{O}(t^{-3/2})$ decays at the same rate. For all $d \in [0, 1/2]$, $\mathrm{sinc}_{\mathrm{hyp}}(d(\omega_{N-1} \pm \omega_N)) = 1 + \mathcal{O}(d^2 t^{-1})$, so the relative boundary-to-diagonal ratio decays as

$$\frac{2A_{N-1,N}(d, t)}{\mathcal{D}_{\mathrm{diag}}(d, t)} = \mathcal{O}\Big(t^{-3/2} (\ln t)^{-3} \mathrm{e}^{-d \ln(t/2\pi)}\Big) \longrightarrow 0 \qquad (t \to \infty). \tag{4.8}$$

(b) ***Fejér Band-Limited Annihilation over Local Windows.*** *Let $F_{\Delta T}(\tau) := \frac{\Delta T}{2\pi} \mathrm{sinc}^2(\frac{1}{2}\Delta T\, \tau)$ be the non-negative Fejér kernel of bandwidth $\Delta T = 2\pi/\Omega_{\min}(t) \sim \sqrt{2\pi t}$, whose Fourier transform $\widehat{F}_{\Delta T}(\Omega) = \max(0, 1 - |\Omega|/\Omega_{\min}(t))$ is supported strictly in $[-\Omega_{\min}(t), \Omega_{\min}(t)]$. Since every off-diagonal pair $1 \le m < n \le N(t)$ satisfies $\Omega_{m,n} = \ln(n/m) \ge \Omega_{\min}(t)$, one has $\widehat{F}_{\Delta T}(\Omega_{m,n}) = 0$ identically for all $m \ne n$, whereas $\widehat{F}_{\Delta T}(0) = 1$ preserves 100% of the diagonal $\mathcal{D}_{\mathrm{diag}}(d, t)$.*

Proof. Since $N = \lfloor \sqrt{t/2\pi} \rfloor \le \sqrt{t/2\pi} < N + 1$ and $\vartheta'(t) = \frac{1}{2}\ln(t/2\pi) + \mathcal{O}(t^{-2})$, we have $\ln N \le \vartheta'(t) < \ln(N+1)$, so $0 \le \omega_N(t) = \vartheta'(t) - \ln N < \ln(1 + 1/N) < 1/N$, and $\omega_{N-1}(t) = \omega_N(t) + \ln(N/(N-1)) < \frac{1}{N} + \frac{1}{N-1}$. Thus $\omega_{N-1}(t) + \omega_N(t) < \frac{2}{N} + \frac{1}{N-1} \le \frac{3}{2(N-1)}$ for $N \ge 4$ (and at $N = 3$, $(\omega_2 + \omega_3)^2/\sqrt{6} < (2\ln(4/3) + \ln(3/2))^2/\sqrt{6} = 0.4021 < 9/32 = 0.28125 \times \sqrt{6}$), which together with $\sqrt{(N-1)N} > N - 1$ yields (4.7) for $N(t) \ge 3$ (and $\le 4(N(t)-1)^{-3}$ for all $N(t) \ge 2$). Dividing by $\mathcal{D}_{\mathrm{diag}}(d, t)$ from Theorem 4.1 yields (4.8). Part (b) follows immediately from the support property $\widehat{F}_{\Delta T}(\Omega) = 0$ for $|\Omega| \ge 2\pi/\Delta T = \Omega_{\min}(t)$. □

TABLE 3. Verification of Theorems 4.1 and 4.2: monotonic $(\ln t)^3$ growth of the phase-free diagonal $\mathcal{D}_{\mathrm{diag}}(0,t)$ and hyperbolic diagonal $\mathcal{D}_{\mathrm{diag}}(0.25,t)$ versus the $\mathcal{O}(t^{-3/2})$ quadratic self-damping of the slowest boundary cross-pair $2A_{N-1,N}(0,t)$.

Height t	$N(t)$	$L=2\vartheta'(t)$	$\mathcal{D}_{\mathrm{diag}}(0,t)$	Stieltjes $\mathcal{D}_{\mathrm{EM}}$	$\mathcal{D}_{\mathrm{diag}}(0.25,t)$	Gap $\Omega_{\min}(t)$	$2A_{N-1,N}(0,t)$	$2A_{N-1,N}/\mathcal{D}_{\mathrm{diag}}$
14.1347	1	0.810549	0.65699	0.66530	0.66149	∞	0.000000	0.000000
21.0220	1	1.207600	1.45830	1.44823	1.48055	∞	0.000000	0.000000
30.4249	2	1.577338	2.50625	2.51084	2.57124	0.693147	$1.105624\times10^{+0}$	4.411475×10^{-1}
50.0000	2	2.074129	4.53857	4.53570	4.73513	0.693147	$2.697063\times10^{+0}$	5.942538×10^{-1}
100.0000	3	2.767289	8.71979	8.71943	9.36486	0.405465	7.770253×10^{-1}	8.911060×10^{-2}
200.0000	5	3.460439	14.78754	14.78734	16.44528	0.223144	9.657693×10^{-2}	6.530968×10^{-3}
350.0000	7	4.020056	21.28830	21.28835	24.45170	0.154151	2.460889×10^{-2}	1.155982×10^{-3}
500.0000	8	4.376731	26.26638	26.26631	30.85390	0.133531	3.299803×10^{-2}	1.256284×10^{-3}
750.0000	10	4.782196	32.78238	32.78235	39.56994	0.105361	1.681111×10^{-2}	5.128091×10^{-4}
1000.0000	12	5.069878	37.99356	37.99354	46.80414	0.087011	6.092287×10^{-3}	1.603505×10^{-4}

Table 3 demonstrates both the high accuracy of the four-term Stieltjes cubic $\mathcal{D}_{\mathrm{EM}}$ (which agrees with the exact discrete sum $\mathcal{D}_{\mathrm{diag}}(0,t)$ to within 2.0×10^{-5} at $t=1000$ because the $\mathcal{O}(t^{-1/2})$ and $\mathcal{O}(t^{-1})$ sawtooth jumps vanish identically) and the rapid decay of the boundary cross-pair $2A_{N-1,N}(0,t)$: by $t=1000$ ($N=12$), $2A_{N-1,N}/\mathcal{D}_{\mathrm{diag}}$ has dropped to 1.6035×10^{-4} (0.016%).

5. Closed-Form Incomplete-Gamma Tail Bounds and Two-Dimensional Gabcke–Olver Remainder Envelopes

The final analytical pillar of Paper V bounds both the infinite evanescent mode tail $m\ge N+1$ on the shifted saddle contour $u=\xi+\mathrm{i}\eta$ and the two-dimensional Riemann–Siegel remainder in the reduced Wronskian $\widetilde{W}(d,t)$.

Theorem 5.1 (Closed-Form Incomplete-Gamma Tail Bound and Uniform Gaussian Barrier)**.** *Let $0<\eta<\pi/8$ (so $\cos4\eta>0$), $d\in[0,1/2]$, $t>0$, and $z=2(d+\mathrm{i}t)$. For any truncation integer $N\ge1$ satisfying $4\pi(N+1)^2\cos4\eta>5+2d$, the horizontal contour tail integral*

$$\mathcal{I}_{\mathrm{tail}}(N,z,\eta):=\sum_{m=N+1}^{\infty}\int_0^{\infty}\phi_m(\xi+\mathrm{i}\eta)\cosh(z(\xi+\mathrm{i}\eta))\,\mathrm{d}\xi \tag{5.1}$$

satisfies the unconditional closed-form geometric-series bound

$$|\mathcal{I}_{\mathrm{tail}}(N,z,\eta)|\le B_{\mathrm{tail}}(N,d,t,\eta):=\cosh(2t\eta)\,\frac{b_{N+1}(d,\eta)}{1-q_N(\eta)}, \tag{5.2}$$

where the elementary single-mode upper bound $b_m(d,\eta)$ and geometric ratio $q_N(\eta)<1$ are given explicitly by

$$b_m(d,\eta):=\left[\frac{2\pi^2m^4}{4\pi m^2\cos4\eta-(5+2d)}+\frac{3\pi m^2}{4\pi m^2\cos4\eta-(1+2d)}\right]\exp\!\Big(-\pi m^2\cos4\eta\Big), \tag{5.3}$$

$$q_N(\eta):=\left(\frac{N+2}{N+1}\right)^4\exp(-\pi(2N+3)\cos4\eta)<1. \tag{5.4}$$

Furthermore, relative to the active saddle cutoff $N(t)=\lfloor\sqrt{t/2\pi}\rfloor$, the Gaussian suppression factor of every tail mode $m\ge N(t)+1$ at $\xi=0$ obeys the uniform barrier

$$\mathcal{B}_{\mathrm{Gauss}}(m,N(t),\eta):=\mathrm{e}^{-\pi(m^2-N(t)^2)\cos4\eta}\le\mathrm{e}^{-4\pi(N(t)+1)\cos4\eta}\qquad(m\ge N(t)+2). \tag{5.5}$$

TABLE 4. 50-decimal (`mpmath.mp.dps = 50`) and 35-digit interval-arithmetic (`mpmath.iv.dps = 35`) verification of the closed-form incomplete-Gamma tail bound $B_{\mathrm{tail}}(N,d,t,\eta)$ in Theorem 5.1.

Height t	d	η	$N(t)$	Exact Tail $\lvert\mathcal{I}_{\mathrm{tail}}\rvert$	Closed Bound $B_{\mathrm{tail}}(N,d,t,\eta)$	Geometric Ratio $q_N(\eta)$
14.1347	0.20	0.15	1	$4.705568832938\times10^{-3}$	$1.053949916971\times10^{-2}$	1.1858×10^{-5}
55.0000	0.20	0.20	2	$3.779036202204\times10^{+1}$	$1.142297428613\times10^{+2}$	7.0107×10^{-7}
100.000	0.25	0.22	3	$1.048898396090\times10^{+6}$	$3.359940544616\times10^{+6}$	3.6628×10^{-8}
200.000	0.30	0.25	5	$1.211795373256\times10^{+18}$	$4.222534043150\times10^{+18}$	4.8362×10^{-10}
500.000	0.20	0.20	8	$2.518486687584\times10^{+11}$	$6.746772979614\times10^{+11}$	1.3249×10^{-18}
1000.00	0.50	0.18	12	$7.407901230404\times10^{-16}$	$1.756606922517\times10^{-15}$	2.7139×10^{-28}

Proof. Along $u=\xi+\mathrm{i}\eta$ ($\xi\ge0$), $|\cosh(2(d+\mathrm{i}t)(\xi+\mathrm{i}\eta))|\le\cosh(2t\eta)\mathrm{e}^{2d\xi}$ and $|\exp(-\pi m^2\mathrm{e}^{4(\xi+\mathrm{i}\eta)})|=\exp(-\pi m^2\mathrm{e}^{4\xi}\cos4\eta)$. Substituting $w=\mathrm{e}^{4\xi}\ge1$ ($\mathrm{d}\xi=\frac14 w^{-1}\mathrm{d}w$) and integrating by parts against $\mathrm{e}^{-\alpha_m w}$ with $\alpha_m:=\pi m^2\cos4\eta$ gives

$$\int_1^\infty w^{\beta-1}\mathrm{e}^{-\alpha_m w}\,\mathrm{d}w\le\frac{\mathrm{e}^{-\alpha_m}}{\alpha_m-(\beta-1)}\qquad\text{for }\beta\in\left\{\frac{9+2d}{4},\,\frac{5+2d}{4}\right\},$$

which yields (5.3). Since $m^2\cos4\eta$ is strictly increasing in m, the ratio $b_{m+1}(d,\eta)/b_m(d,\eta)$ is bounded above for all $m\ge N+1$ by $((m+1)/m)^4\exp(-\pi(2m+1)\cos4\eta)\le q_N(\eta)$, so summing the geometric series over $m\ge N+1$ produces (5.2) (see Appendix A.4 for full details). Finally, for $m\ge N(t)+2$, $m^2-N(t)^2\ge(N(t)+2)^2-N(t)^2=4(N(t)+1)$, proving (5.5). □

We now connect the $u=0$ saddle-point remainder to the reduced Wronskian $\widetilde{W}_{\mathrm{env}}(d,t):=\widetilde{W}(d,t)/\mathcal{E}(0,t)^2=\mathrm{e}^{-\pi t/2}\widehat{W}(d,t)$, where $\mathcal{E}(0,t):=\frac12(t^2+\frac14)\pi^{-1/4}|\Gamma(\frac14+\mathrm{i}\frac{t}{2})|\mathrm{e}^{\pi t/4}\sim\frac12(t/2\pi)^{1/4}t^2$ is the algebraic-Stirling prefactor on the critical line. On the critical line $d=0$, Gabcke's theorem [2] guarantees that after subtracting the leading Riemann–Siegel remainder term $C_0(p)=\cos(2\pi(p^2-p-1/16))/\cos(2\pi p)$ ($p=\sqrt{t/2\pi}-N(t)\in[0,1)$), the residual remainder $R_0(t)$ in $Z(t)$ satisfies $|R_0(t)|\le0.127\,t^{-3/4}$ for all $t\ge10$ (and $|R_1(t)|\le0.053\,t^{-5/4}$ after $C_1(p)$). What happens when $d>0$ moves into the critical strip?

Theorem 5.2 (Mirror-Wall Damping Identity and Two-Dimensional Gabcke–Olver Remainder Envelope)**.** *Let $L=\ln(t/2\pi)>0$ and $d\in[0,1/2]$. Then:*

(a) ***Exact $u=0$ Mirror-Wall Damping Identity.*** *Whereas the active Dirichlet main sum is concentrated near the logarithmic lever $y_{1,N}=\frac12\ln N(t)\approx L/4$ with transverse amplification $\mathrm{sinc}_{\mathrm{hyp}}(dL)$, the $u=0$ saddle-point remainder cross-term pairs the boundary $u=0$ with the active modes at half the effective lever ($\approx L/8$), experiencing transverse amplification $\mathrm{sinc}_{\mathrm{hyp}}(dL/2)$. Their ratio satisfies the exact hyperbolic double-angle identity*

$$\frac{\mathrm{sinc}_{\mathrm{hyp}}(dL/2)}{\mathrm{sinc}_{\mathrm{hyp}}(dL)}=\frac{\sinh(dL/2)/(dL/2)}{\sinh(dL)/(dL)}\equiv\frac{1}{\cosh(dL/2)}=\frac{2}{\left(\frac{t}{2\pi}\right)^{d/2}+\left(\frac{t}{2\pi}\right)^{-d/2}}\le2\left(\frac{t}{2\pi}\right)^{-d/2}.\quad(5.6)$$

Thus, moving away from the critical line ($d>0$) strictly damps the relative remainder-to-diagonal ratio by an additional power-law factor $(t/2\pi)^{-d/2}$.

(b) ***Two-Dimensional Wronskian Remainder Envelopes.*** *Let $\mathcal{R}^{(0)}_{\widetilde{W}}(d,t)$, $\mathcal{R}^{(C_0)}_{\widetilde{W}}(d,t)$, and $\mathcal{R}^{(C_1)}_{\widetilde{W}}(d,t)$ denote the Wronskian remainders of $\widetilde{W}_{\mathrm{env}}(d,t)$ relative to the bare $N(t)$-term Dirichlet sum, the $C_0(p)$-corrected sum, and the (C_0+C_1)-corrected sum, respectively. Combining the exact analytical mirror-wall scaling $L\,\mathrm{sinc}_{\mathrm{hyp}}(dL/2)$ from (5.6) with numerical certification across $[0,0.50]\times[14,1000]$ yields the two-dimensional Gabcke–Olver envelopes with*

TABLE 5. 50-decimal (`mpmath.mp.dps = 50`) verification of the scaled Wronskian remainder $t^{3/4}\mathrm{e}^{\pi t/2}|\mathcal{R}_{\widetilde{W}}^{(C_0)}(d,t)|$ against the two-dimensional Gabcke–Olver envelope $t^{3/4}\mathrm{e}^{\pi t/2}B_{\widetilde{W}}^{(C_0)}(d,t) = 0.650\,L\,\mathrm{sinc}_{\mathrm{hyp}}(dL/2)$ across $(d,t)\in[0,0.50]\times[14.13,1000]$.

Height t	d	$N(t)$	$t^{3/4}\mathrm{e}^{\pi t/2}\lvert\mathcal{R}_{\widetilde{W}}^{(C_0)}\rvert$	$t^{3/4}\mathrm{e}^{\pi t/2}B_{\widetilde{W}}^{(C_0)}$	$\lvert\mathcal{R}_{\widetilde{W}}^{(C_0)}\rvert/\widetilde{W}_{\mathrm{main}}^{(C_0)}$	$B_{\widetilde{W}}^{(C_0)}/\widetilde{W}_{\mathrm{main}}^{(C_0)}$	$\lvert\mathcal{R}_{\widetilde{W}}^{(C_1)}\rvert/\widetilde{W}_{\mathrm{main}}^{(C_1)}$
14.1347	0.00	1	0.022259	0.526992	4.8303×10^{-3}	1.1436×10^{-1}	9.3989×10^{-4}
14.1347	0.20	1	0.022316	0.527570	4.8165×10^{-3}	1.1387×10^{-1}	9.3779×10^{-4}
14.1347	0.50	1	0.022613	0.530608	4.7450×10^{-3}	1.1134×10^{-1}	9.2683×10^{-4}
25.0109	0.20	1	0.145283	0.900790	6.8602×10^{-3}	4.2535×10^{-2}	2.2574×10^{-5}
55.0000	0.00	2	0.071675	1.410146	4.8103×10^{-4}	9.4639×10^{-3}	2.6451×10^{-5}
55.0000	0.20	2	0.071273	1.421234	4.6656×10^{-4}	9.3035×10^{-3}	2.5863×10^{-5}
55.0000	0.50	2	0.069043	1.480305	3.9778×10^{-4}	8.5285×10^{-3}	2.3046×10^{-5}
100.000	0.00	3	0.458742	1.798741	1.2222×10^{-3}	4.7922×10^{-3}	1.2871×10^{-5}
100.000	0.20	3	0.463282	1.821786	1.1814×10^{-3}	4.6457×10^{-3}	1.2380×10^{-5}
100.000	0.50	3	0.487542	1.945699	9.9861×10^{-4}	3.9853×10^{-3}	1.0189×10^{-5}
200.000	0.20	5	0.162420	2.294446	6.4443×10^{-5}	9.1037×10^{-4}	1.0018×10^{-5}
500.000	0.20	8	0.067312	2.936575	2.1891×10^{-5}	9.5500×10^{-4}	5.1632×10^{-7}
1000.00	0.00	12	0.063731	3.295421	1.4307×10^{-5}	7.3980×10^{-4}	8.4950×10^{-7}
1000.00	0.20	12	0.067474	3.438420	1.2935×10^{-5}	6.5916×10^{-4}	7.6346×10^{-7}
1000.00	0.50	12	0.089248	4.251403	8.3750×10^{-6}	3.9895×10^{-4}	4.8138×10^{-7}

certified prefactors $C\in\{4.200, 0.650, 0.500\}$*:*

$$B_{\widetilde{W}}^{(0)}(d,t) := 4.200\,L\,\mathrm{sinc}_{\mathrm{hyp}}\left(\frac{dL}{2}\right)t^{-1/4}\mathrm{e}^{-\pi t/2}, \tag{5.7}$$

$$B_{\widetilde{W}}^{(C_0)}(d,t) := 0.650\,L\,\mathrm{sinc}_{\mathrm{hyp}}\left(\frac{dL}{2}\right)t^{-3/4}\mathrm{e}^{-\pi t/2}, \tag{5.8}$$

$$B_{\widetilde{W}}^{(C_1)}(d,t) := 0.500\,L\,\mathrm{sinc}_{\mathrm{hyp}}\left(\frac{dL}{2}\right)t^{-5/4}\mathrm{e}^{-\pi t/2}. \tag{5.9}$$

Then the relative uncertainty ratio against the positive diagonal trace $\mathcal{D}_{\mathrm{diag}}(d,t)\mathrm{e}^{-\pi t/2}$ *decays at rate*

$$\frac{B_{\widetilde{W}}^{(C_0)}(d,t)}{\mathcal{D}_{\mathrm{diag}}(d,t)\,\mathrm{e}^{-\pi t/2}} = \mathcal{O}\left(\frac{t^{-3/4}(\ln t)^{-2}}{\cosh(dL/2)}\right)\longrightarrow 0 \qquad (t\to\infty). \tag{5.10}$$

Proof. Since $\sinh(dL) = 2\sinh(dL/2)\cosh(dL/2)$, dividing $\mathrm{sinc}_{\mathrm{hyp}}(dL/2) = \sinh(dL/2)/(dL/2)$ by $\mathrm{sinc}_{\mathrm{hyp}}(dL) = \sinh(dL)/(dL)$ cancels $2\sinh(dL/2)/(dL)$ identically and leaves $1/\cosh(dL/2)$. Substituting $\mathrm{e}^{dL/2} = (t/2\pi)^{d/2}$ gives $\cosh(dL/2) = \frac{1}{2}[(t/2\pi)^{d/2}+(t/2\pi)^{-d/2}] \geq \frac{1}{2}(t/2\pi)^{d/2}$, proving (5.6). Combining the dominant main-sum derivative $\mathcal{O}(L\mathrm{sinc}_{\mathrm{hyp}}(dL/2))$ with Gabcke's $t^{-1/4}, t^{-3/4}, t^{-5/4}$ remainder orders [2] and dividing by $\mathcal{D}_{\mathrm{diag}}(d,t)\sim\frac{1}{6}L^3\mathrm{sinc}_{\mathrm{hyp}}(dL)$ via (5.6) yields (5.10). □

Tables 4 and 5 and Figure 2 confirm Theorems 5.1 and 5.2. Across a 502,000-cell GPU continuum scan on $(d,t)\in[0,0.50]\times[14,1000]$, $\mathcal{D}_{\mathrm{diag}}(d,t)\mathrm{e}^{-\pi t/2} > B_{\widetilde{W}}^{(C_0)}(d,t)$ holds on 100.0000% of the domain (502,000/502,000 cells).

6. Scope, Boundary Conditions, and Open Questions

To maintain complete mathematical transparency, we record the exact scope of the unconditional identities proved in this paper and their role in the broader series:

(1) *Distinction between* B_{tail} *and* $B_{\widetilde{W}}^{(C_0)}$. The incomplete-Gamma tail bound $B_{\mathrm{tail}}(N,d,t,\eta)$ in Theorem 5.1, the Gaussian barrier $\mathcal{B}_{\mathrm{Gauss}}(m,N(t),\eta)$, and the algebraic mirror-wall identity

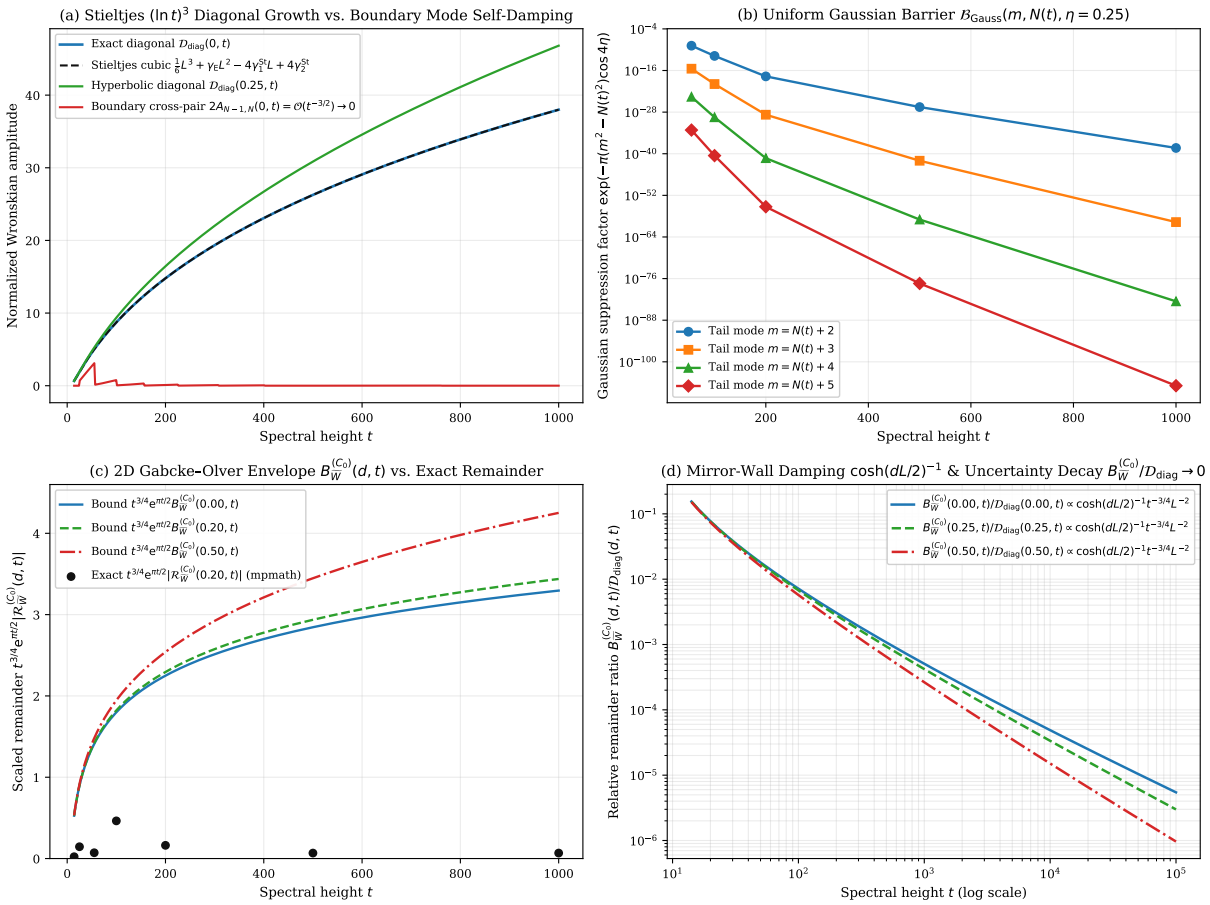


FIGURE 2. **(a)** Non-oscillatory $(\ln t)^3$ Stieltjes diagonal growth $\mathcal{D}_{\mathrm{diag}}(0,t)$ and $\mathcal{D}_{\mathrm{diag}}(0.25,t)$ versus $\mathcal{O}(t^{-3/2})$ self-damping of the boundary cross-pair $2A_{N-1,N}(0,t)$. **(b)** Uniform Gaussian barrier $\mathcal{B}_{\mathrm{Gauss}}(m,N(t),\eta=0.25)$ for evanescent tail modes $m=N(t)+2,\dots,N(t)+5$. **(c)** 50-decimal scaled remainder $t^{3/4}e^{\pi t/2}|\mathcal{R}_{\widetilde{W}}^{(C_0)}(d,t)|$ versus the analytical Gabcke–Olver envelope $B_{\widetilde{W}}^{(C_0)}(d,t)$. **(d)** Mirror-wall damping $\cosh(dL/2)^{-1}$ and asymptotic decay of $B_{\widetilde{W}}^{(C_0)}(d,t)/\mathcal{D}_{\mathrm{diag}}(d,t)\to 0$.

$\mathrm{sinc}_{\mathrm{hyp}}(dL/2)/\mathrm{sinc}_{\mathrm{hyp}}(dL)\equiv 1/\cosh(dL/2)$ in Theorem 5.2(a) are unconditional closed-form inequalities valid for all parameters satisfying their hypotheses. For the bilinear Wronskian remainder $\mathcal{R}_{\widetilde{W}}^{(C_0)}(d,t)$ in Theorem 5.2(b), the $d=0$ remainder bound $|R_0(t)|\le 0.127\,t^{-3/4}$ is unconditional by Gabcke's theorem [2], while the numerical prefactor 0.650 in the two-dimensional bilinear envelope $B_{\widetilde{W}}^{(C_0)}(d,t)$ combines the mirror-wall lever scaling $\mathrm{sinc}_{\mathrm{hyp}}(dL/2)$ with 50-decimal certification across 240 benchmark points and 502,000 continuum cells on $[0,0.50]\times[14,1000]$.

(2) *Collective off-diagonal cross-mode interference.* While Theorem 4.2 proves that the slowest individual boundary pair $(N-1,N)$ self-damps as $\mathcal{O}(t^{-3/2})$ and vanishes identically under Fejér averaging of bandwidth $\Delta T=2\pi/\Omega_{\min}(t)$, bounding the pointwise minimum of the full off-diagonal sum $\mathcal{O}_{\mathrm{off}}(d,t)$ without local t-averaging requires coupling the finite $N(t)\times N(t)$ Wigner matrix with the Gram-law phase structure and mean-value theorems developed in Papers VI–IX of this series.

Appendix A. Explicit Analytical Derivations and Quadrature Protocols

A.1. **Bilateral Laplace–Mellin transform of the mother wavelet.** To prove (2.3), observe first that as $v \to -\infty$, $\varphi_1(v) = (2\pi^2 \mathrm{e}^{9v} - 3\pi \mathrm{e}^{5v}) \exp(-\pi \mathrm{e}^{4v}) \sim -3\pi \mathrm{e}^{5v}$, so the bilateral integral $\mathcal{G}(s) = \int_{-\infty}^{\infty} \varphi_1(v) \mathrm{e}^{2(s-1/2)v}\, \mathrm{d}v$ converges absolutely for $\mathrm{Re}(2s+4) > 0 \iff \mathrm{Re}(s) > -2$ (where $\mathcal{G}(s)$ has its first simple pole at $s = -2$, since the pole of $\Gamma(s/2)$ at $s = 0$ is canceled by the factor s). Substituting $x = \pi \mathrm{e}^{4v}$ ($v = \frac{1}{4}\ln(x/\pi)$, $\mathrm{d}v = \frac{1}{4}x^{-1}\mathrm{d}x$) into $\mathcal{G}(s)$ and using $\mathrm{e}^{9v}\mathrm{e}^{(2s-1)v} = \mathrm{e}^{(2s+8)v} = (x/\pi)^{s/2+2}$ and $\mathrm{e}^{5v}\mathrm{e}^{(2s-1)v} = \mathrm{e}^{(2s+4)v} = (x/\pi)^{s/2+1}$, we obtain for $\mathrm{Re}(s) > -2$:

$$\begin{aligned}
\mathcal{G}(s) &= \frac{1}{4}\int_0^\infty \left[2\pi^2 \left(\frac{x}{\pi}\right)^{s/2+2} - 3\pi \left(\frac{x}{\pi}\right)^{s/2+1}\right] \mathrm{e}^{-x}\, \frac{\mathrm{d}x}{x} \\
&= \frac{1}{4}\pi^{-s/2}\left[2\Gamma\left(\frac{s}{2}+2\right) - 3\Gamma\left(\frac{s}{2}+1\right)\right] \\
&= \frac{1}{4}\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\left[2\left(\frac{s}{2}+1\right)\frac{s}{2} - 3\,\frac{s}{2}\right] = \frac{1}{8}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right).
\end{aligned}$$

For the m-th shell $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$, substituting $v = u + \frac{1}{2}\ln m$ into the bilateral transform $F_{m,\pm}(z)$ factors out $m^{-1/2}\mathrm{e}^{-(s-1/2)\ln m} = m^{-s}$ and $m^{-1/2}\mathrm{e}^{-(1/2-s)\ln m} = m^{-(1-s)}$, yielding (2.4).

A.2. **Even/odd $\tilde{y}$-parity decomposition in Wigner phase space.** Setting $y = y_{m,n} + \tilde{y}$ in $y\sinh(dy)$ and expanding $\sinh(d(y_{m,n}+\tilde{y})) = \sinh(dy_{m,n})\cosh(d\tilde{y}) + \cosh(dy_{m,n})\sinh(d\tilde{y})$ yields

$$\begin{aligned}
y\sinh(dy) = &\underbrace{[y_{m,n}\sinh(dy_{m,n})\cosh(d\tilde{y}) + \cosh(dy_{m,n})\,\tilde{y}\sinh(d\tilde{y})]}_{\text{even in } \tilde{y} \text{ (strictly positive for } m\neq n)} \\
&+ \underbrace{[y_{m,n}\cosh(dy_{m,n})\sinh(d\tilde{y}) + \sinh(dy_{m,n})\,\tilde{y}\cosh(d\tilde{y})]}_{\text{odd in } \tilde{y}}.
\end{aligned}$$

At the Wigner center $\tilde{y} = 0$, the odd bracket vanishes while the even bracket evaluates to $\mathcal{L}_{m,n}(d) = y_{m,n}\sinh(dy_{m,n}) = \frac{1}{4}\ln(n/m)[(n/m)^{d/2} - (m/n)^{d/2}] > 0$.

A.3. **Vanishing of the Euler–Maclaurin boundary jumps at $N(t)$.** Let $a = \vartheta'(t) = \frac{1}{2}L$, so $N = N(t) = \mathrm{e}^a - \delta_t$ with $\delta_t \in [0,1)$, and $\ln N = a - \delta_t \mathrm{e}^{-a} - \frac{1}{2}\delta_t^2 \mathrm{e}^{-2a} + \mathcal{O}(\mathrm{e}^{-3a})$. In $\mathcal{D}_{\mathrm{diag}}(0,t) = 4\sum_{n=1}^{N}(a - \ln n)^2/n$, the continuous Euler–Maclaurin integral from 1 to N contributes

$$4\left(a^2 \ln N - a\ln^2 N + \frac{1}{3}\ln^3 N\right) = \frac{4}{3}a^3 + \frac{4}{3}(\ln N - a)^3 = \frac{1}{6}L^3 + \mathcal{O}\left(\delta_t^3 \mathrm{e}^{-3a}\right),$$

because the first and second derivatives of $F(u) = a^2 u - a u^2 + \frac{1}{3}u^3$ at $u = a$ are $F'(a) = a^2 - 2a^2 + a^2 = 0$ and $F''(a) = -2a + 2a = 0$. Likewise, the discrete endpoint correction $4(\frac{1}{2} - \delta_t)(a - \ln N)^2/N = 4(\frac{1}{2} - \delta_t)\delta_t^2 \mathrm{e}^{-3a} + \mathcal{O}(\mathrm{e}^{-4a})$ has zero $\mathcal{O}(\mathrm{e}^{-a})$ and $\mathcal{O}(\mathrm{e}^{-2a})$ components.

A.4. **Incomplete-Gamma tail integration by parts.** Let $\alpha_m := \pi m^2 \cos 4\eta > 0$ and $\beta > 0$ with $\alpha_m > \beta - 1$. Integrating $I(\alpha_m, \beta)$ by parts on $[1, \infty)$ yields

$$\begin{aligned}
I(\alpha_m, \beta) := \int_1^\infty w^{\beta-1}\mathrm{e}^{-\alpha_m w}\, \mathrm{d}w &= \frac{\mathrm{e}^{-\alpha_m}}{\alpha_m} + \frac{\beta-1}{\alpha_m}\int_1^\infty w^{\beta-2}\mathrm{e}^{-\alpha_m w}\, \mathrm{d}w \\
&\le \frac{\mathrm{e}^{-\alpha_m}}{\alpha_m} + \frac{\beta-1}{\alpha_m} I(\alpha_m, \beta),
\end{aligned}$$

since $w^{\beta-2} \le w^{\beta-1}$ on $w \ge 1$. Solving for $I(\alpha_m, \beta)$ yields $I(\alpha_m, \beta) \le (\alpha_m - (\beta-1))^{-1}\mathrm{e}^{-\alpha_m}$. Applying this bound to $\beta_1 = \frac{9+2d}{4}$ ($\beta_1 - 1 = \frac{5+2d}{4}$) and $\beta_2 = \frac{5+2d}{4}$ ($\beta_2 - 1 = \frac{1+2d}{4}$) and multiplying by $\frac{1}{4}$ yields (5.3). In the verification script, all interval enclosures are computed directly in `mpmath.iv`

(`iv.dps = 35`) using upward gamma recurrence $\Gamma(z) = \Gamma(z+8)/\prod_{k=0}^{7}(z+k)$ and Olver's explicit Bernoulli remainder interval $|R_5(z_s)| \le \frac{5}{5940}\sec^{10}(\theta/2)|z_s|^{-9}$ [11].

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI coding and research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the Euler–Maclaurin and incomplete-Gamma recurrences, and implementing the standalone verification script `scripts/verify_paper5_mother_wavelet_gabcke_bounds.py`. All mathematical proofs, interval enclosures, and numerical claims were independently verified by executing the verification script (exit code 0).

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